SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1510

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 20

Concept Mapping Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

Code of Conduct

I Prabhavathi R (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Prabhavathi R

Concept Map 1 : Cloud Computing Fundamentals

Cloud Computing

Characteristics

On-demand Access
* Resources are available whenever needed

Resource pooling
* Resource are shared among multiple users

Elasticity
* Resources can handle growth in demand seamlessly

Measured Service
* usage is monitored, controlled and billed as per use

Benefits

Cost Efficiency
pay-as-you-go module reduces cost

High availability
services are available 24/7 with reliability

Flexibility
Resources can be changed as per business need

Global Access
Access services from anywhere in the world

Limits

Security issues
Data may be vulnerable to attacks

Privacy Concerns
Data stored outside organizational control

Limitations

Privacy concerns
Data stored outside Organization control

Security Issues
Data stored outside organizational control

Internet Dependency
Requires stable Internet connection

Vendor Lock-in
Moving from one cloud provider to another can be different.

Concept Map 2 : Evolution of cloud computing

Hardware Evolution

* Mainframes
* Personal Computers
* Servers
* Virtualization technology

Leads to

Efficient use of hardware
better resources utilization
through virtualization

Internet & software Evolution

* Internet growth
* Web application
* APIs and web services

enable

Easy access, connectivity and development of internet based services

Distributed System

* Grid Computing
* cluster computing
* utility computing
* Peer to peer systems

Enable

Resource sharing
scalibility
Fault tolerance

Business needs

* Cost reduction
* Agility
* Flexibility
* Global Reach

drives

Demand for on-demand services and outsourcing of IT resources

Cloud Computing Evolution

results in

Security Delivery Models

Iaas (Infrastructure as a service)

Paas (Platform as a service)

Saas (Software as a service)

Xaas (Everything as a services)

Concept Map 3: Service Models

Cloud Service Models

IaaS
(Infrastructure as a service)
→ Provides
→ Infrastructure Layer (servers, storage, Network, Virtual machines)
→ Provider Responsibilities
Manage physical Infrastructure, servers, storage, networking, virtualization.
→ User Responsibilities

PaaS
(Platform as a service)
→ provides
→ Platform Layer (OS, Runtime, Database, Development tools)
→ Provider Responsibilities
Manages Infrastructure, OS, runtime, middleware, database, tools
→ User Responsibilities

SaaS
(Software as a service)
→ provides
→ Application Layer (Complete software Applications)
→ Provider Responsibilities
Manages everything (infrastructure, platform and applications)
→ User Responsibilities

XaaS
(Everything as a service)
→ provides
→ Any service delivered as a service over cloud
→ Provider responsibilities
Delivers and manages the service
→ User Responsibilities

Shared Responsibility Model
→ Provider (cloud service Provider)
→ Consumer (cloud user)

Concept Map 4: Virtualization and Web services as Enablers of cloud

Enabled by

Virtualization

- Creates virtual machines (VMs)
- ↓
- VMs run on Hypervisor
- ↓
- Hypervisor provides abstraction of hardware.
- ↓
- Enables Resource Pooling and Multi-tenancy
- ↓
- Supports Scalability, Elasticity and Flexibility
- ↓
- Reduces cost and Improves Resource utilization

Web services

- use APIs for Communication
- ↓
- APIs Implemented using
  - SOAP
  - REST
- Connects Provider and Consumer Over Internet
- ↓
- Enable Interoperability and Loose Coupling
- ↓
- Facilitate On-demand service delivery

Virtualization → enables → **Cloud Computing** ← enables ← Web services

Cloud Computing → Together Enables →

Cloud Deployment and operations

- * On-demand Access
- * Elasticity
- * Measured service
- * Scalability
- * Resource pooling
- * High Availability